Do the stances of Christian churches influence identifiers' politcial beliefs?

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1 Original Study

The study I chose to replicate for this project: "The Influence of Religious–Political Sophistication on U.S. Public Opinion" by Eric Schmidt (2017). The main purpose of this study is to discover whether and how religious institutes' viewpoints affect parishioners' understanding and reasoning on political issues. This phenomenon is referred to as Religious-Political Sophistication (RPS). Other literature on this topic, discuss the direct influence of religious leaders and how these teachings potentially influence religious identifiers' political beliefs/allignment. Schmidt suggests on top of direct influence, religious identifiers also take in account their church's stance on issues (i.e. the Roman Catholic Church's stance on abortion). Therefore, the author is looking to test three hypotheses regarding RPS:

- Religious identifiers who know their church's teachings are more likely to share these views
- RPS has a greater impact on their public opinion in comparison to secular knowledge
- RPS helps connect religious identifers' faith to politics

Method

To test the hypotheses above, the author took data from 2014 Cooperative Congressional Election Study (CCES) and an original CCES module. From there, he took the best represented religious traditions (Roman Catholic, Evangelical Protestants, and Mainline Protestants) to analyze their political positions on same sex marriage and abortion.

2 Data

```r
#insert data
rep_data <-  read.csv("/Users/carolinekajkowski/Desktop/Trinity/StatsII_2026/
   replication/my_answers/original_files/replication-data.csv")

#subsetting the data to exclude ambiguous answers
mainliners <- subset(rep_data, mainline == 1 & ambiguous == 0)
evan.cath <- subset(rep_data, mainline != 1 & ambiguous==0)
```

In the original R script, the author decides to separate the dataset to create two new datasets with all inputs of Roman Catholic and Evangelical Protestants in one and Mainline Protestants in another. The justification for this in the paper was that the author found that the Roman Catholic church and Evangelical Protestants have more clear universal opinions on same sex marriage and abortion rights, while Mainline Protestants have varrying views on these social policies. Therefore, the study is focused on the effect of RPS on Roman Catholic and Evangelical Protestants while treating Mainline Protestants as the baseline to compare later on.

Input variables:

Party ID (0 = strong Democrat; 6 = strong Republican)

Church attendance (0 = never; 5 = more than once per week)

Biblical Literacy (the number of correct answers to three multiple-choice questions about the Bible's books and figures)

Secular Political Knowledge (a three-point scale indicating each respondent's number of correct responses to two questions about national politics)

RPS (1 =respondent knows official denomination stance; 0 = respondent doesn't know)

Controled variables: age, ideology, gender, race, education, family income, and religious tradition (1 = Roman Catholic; 0 = Evangelical Protestant)

Outcome variable:

Support (1 = support rights (same-sex marriage or abortion); 0 = oppose)

3 Original Models

The author creates binomial logistic regression models that are used throughout the paper:

Secular Model:

$$\begin{aligned}\text{Support} = \beta_0 + \beta_1\,\text{Party Identification} + \beta_2\,\text{Secular Political Knowledge}\\ + \beta_3\,(\text{Party Identification} \times \text{Secular Political Knowledge})\end{aligned}$$

Naive Model:

$$\begin{aligned}\text{Support} = \beta_0 &+ \beta_1\,\text{Party Identification} + \beta_2\,\text{Secular Political Knowledge}\\ &+ \beta_3\,(\text{Party Identification} \times \text{Secular Political Knowledge})\\ &+ \beta_4\,\text{Church Attendance} + \beta_5\,\text{Biblical Literacy}\\ &+ \beta_6\,(\text{Church Attendance} \times \text{Biblical Literacy})\end{aligned}$$

"Culture Wars" Model:

$$\begin{aligned}\text{Support} = \beta_0 &+ \beta_1\,\text{Party Identification} + \beta_2\,\text{Secular Political Knowledge}\\ &+ \beta_3\,(\text{Party Identification} \times \text{Secular Political Knowledge})\\ &+ \beta_4\,\text{Church Attendance} + \beta_5\,\text{Biblical Literacy}\\ &+ \beta_6\,(\text{Church Attendance} \times \text{Biblical Literacy})\\ &+ \beta_7\,(\text{Church Attendance} \times \text{Secular Political Knowledge})\end{aligned}$$

RPS Model:

$$\begin{aligned}\text{Support} = \beta_0 &+ \beta_1\,\text{Party Identification} + \beta_2\,\text{Secular Political Knowledge}\\ &+ \beta_3\,(\text{Party Identification} \times \text{Secular Political Knowledge})\\ &+ \beta_4\,\text{Church Attendance} + \beta_5\,\text{Biblical Literacy}\\ &+ \beta_6\,(\text{Church Attendance} \times \text{Biblical Literacy})\\ &+ \beta_7\,(\text{Church Attendance} \times \text{Secular Political Knowledge})\\ &+ \beta_8\,\text{RPS}\\ &+ \beta_9\,(\text{Church Attendance} \times \text{RPS})\end{aligned}$$

Tables 1 and 2 in this paper look at Evangelical Protestants and Roman Catholics support of same-sex marriage and abortion rights (respectively) using all four models above. The author concludes that the secular models provides insight

Table 3 takes a different approach and looks at whether the respondent's stance on same sex marriage predicts their stance on abortion. So the model looks a bit different (shown in the code below)

```
#table 3: Linking of views on ssm and abortion (Roman Catholics &
    Evangelical Protestants)
model1 = glm(ssm ~ age + female + black + education.collapsed
           + income.collapsed + ideology.collapsed + romancatholic
           + partyid + secular.knowledge + partyid*secular.knowledge
           + church + bible.collapsed + church*bible.collapsed
           + church*secular.knowledge + rps.abortion
           + church*rps.abortion, data=evan.cath, family="binomial")
summary(model1)

model2 = glm(abortion ~ age + female + black + education.collapsed
           + income.collapsed + ideology.collapsed + romancatholic
           + partyid + secular.knowledge + partyid*secular.knowledge
```

```
            + church + bible.collapsed + church*bible.collapsed
            + church*secular.knowledge + rps.ssm
            + church*rps.ssm, data=evan.cath, family="binomial")
summary(model2)
```

Tables 4 and 5 look at the Mainline Protestants. As mentioned before, the data was split into two datasets because the author believed that Mainline protestants have more mixed views on this topics in comparison to Roman Catholics and Evangelical Protestants. The author also chose to not include all the demographic variables in the Mainline Protestant models.

Tables 6 and 7

```
#Data manipulation for 6&7
# Reports of Church teaching (ssm)
unique(rep_data$church.ssm)
rep_data$church.ssm <- as.character(rep_data$church.ssm)
str(rep_data$church.ssm)
church.reports.ssm <- recode(rep_data$church.ssm,
                             "Not Available" = NA_character_)
table(church.reports.ssm)

# Reports of Church Teaching (Abortion)
# Take the 'Not Available' out
unique(rep_data$church.abortion)
table(rep_data$church.abortion)
str(rep_data$church.abortion)
church.reports.abortion <- recode(rep_data$church.abortion,
                                  "Not Available" = NA_character_)
table(church.reports.abortion)
#------------------------------------------------------------

#table 6: RPS answer (SSM) and support/opposition to SSM
table6 <-  prop.table(table(church.reports.ssm[rep_data$ambiguous==0], rep_data$ssm[rep_data$ambiguous==0]), 2)
round(table6, digits = 2)
table6_counts <- table(church.reports.ssm[rep_data$ambiguous==0], rep_data$ssm[rep_data$ambiguous==0])
sum(table6_counts[, 1])
sum(table6_counts[, 2])

stargazer(table6,
          type = "latex",
          summary = FALSE,
          rownames = TRUE,
          title = "Beliefs about church teaching on same-sex marriage by personal opinion on same-sex marriage Replication",
          digits = 2)
#------------------------------------------------------------

#table 7: RPS answer (abortion) and support/opposition
table7 <-  prop.table(table(church.reports.abortion[rep_data$ambiguous==0], rep_data$abortion[rep_data$ambiguous==0]), 2)
```

```
round(table7, digits = 2)
table7_counts <- table(church.reports.abortion[rep_data$ambiguous==0], rep_data$abortion[rep_data$ambiguous==0])
sum(table7_counts[, 1])
sum(table7_counts[, 2])

stargazer(table7,
          type = "latex",
          summary = FALSE,
          rownames = TRUE,
          title = "Beliefs about church teaching on abortion by personal opinion on unrestricted abortion rights Replication",
          digits = 2)
```

Table 8

```
inc.ssm <- glm(ssm ~ age + female + black + education.collapsed
               + income.collapsed + ideology.collapsed + romancatholic
               + partyid + secular.knowledge + partyid*secular.knowledge
               + church + bible.collapsed + church*bible.collapsed
               + church*secular.knowledge + incorrect.rps.ssm
               + church*incorrect.rps.ssm, data=evan.cath, family="binomial")
summary(inc.ssm)


inc.abortion <- glm(abortion ~ age + female + black + education.collapsed
                    + income.collapsed + ideology.collapsed + romancatholic
                    + partyid + secular.knowledge + partyid*secular.knowledge
                    + church + bible.collapsed + church*bible.collapsed
                    + church*secular.knowledge + incorrect.rps.abortion
                    + church*incorrect.rps.abortion,
                    data=evan.cath, family="binomial")
summary(inc.abortion)
```

Table 1: Evangelical Protestants and Roman Catholics' support of SSM Replication

	Dependent variable:			
	ssm			
	Secular	Naive	Culture Wars	RPS
	(1)	(2)	(3)	(4)
age	−0.025**	−0.025**	−0.025**	−0.032**
	(0.011)	(0.013)	(0.013)	(0.015)
female	0.533*	0.481	0.478	0.726*
	(0.323)	(0.351)	(0.352)	(0.400)
black	0.059	−0.138	−0.140	−0.331
	(0.493)	(0.586)	(0.586)	(0.633)
education.collapsed	0.055	0.075	0.076	0.014
	(0.126)	(0.144)	(0.144)	(0.165)
income.collapsed	0.013	0.037	0.038	0.048
	(0.058)	(0.063)	(0.064)	(0.072)
ideology.collapsed	−0.830***	−0.793***	−0.795***	−0.922***
	(0.250)	(0.281)	(0.281)	(0.313)
romancatholic	1.571***	1.303***	1.309***	1.111***
	(0.336)	(0.366)	(0.371)	(0.409)
partyid	−0.249*	−0.242	−0.245	−0.183
	(0.143)	(0.163)	(0.165)	(0.197)
secular.knowledge	−0.082	−0.141	−0.108	0.040
	(0.308)	(0.354)	(0.471)	(0.519)
church		−0.080	−0.061	0.044
		(0.263)	(0.322)	(0.378)
bible.collapsed		−0.262	−0.261	−0.084
		(0.512)	(0.511)	(0.550)
rps.ssm				−0.596
				(0.764)
partyid:secular.knowledge	−0.002	0.002	0.004	−0.078
	(0.088)	(0.098)	(0.099)	(0.116)
church:bible.collapsed		−0.074	−0.074	−0.054
		(0.171)	(0.170)	(0.185)
secular.knowledge:church			−0.014	0.075
			(0.138)	(0.159)
church:rps.ssm				−0.605**
				(0.262)
Constant	1.848**	2.848**	2.800**	3.446**
	(0.834)	(1.148)	(1.236)	(1.384)
Observations	280	246	246	246
Log Likelihood	−139.734	−118.555	−118.549	−97.484
Akaike Inf. Crit.	301.468	265.110	267.099	228.969
Note:				*p<0.1; **p<0.05; ***p<0.01

Table 2: Evangelical Protestants and Roman Catholics' support of abortion Replication

	Dependent variable:			
	abortion			
	Secular	Naive	Culture Wars	RPS
	(1)	(2)	(3)	(4)
age	−0.012	−0.019	−0.019	−0.019
	(0.011)	(0.013)	(0.013)	(0.013)
female	0.419	0.298	0.297	0.319
	(0.326)	(0.363)	(0.364)	(0.371)
black	0.411	−0.085	−0.087	−0.150
	(0.505)	(0.629)	(0.631)	(0.629)
education.collapsed	−0.069	−0.042	−0.042	−0.067
	(0.128)	(0.150)	(0.151)	(0.155)
income.collapsed	0.077	0.036	0.036	0.036
	(0.059)	(0.065)	(0.065)	(0.067)
ideology.collapsed	−1.120***	−1.320***	−1.320***	−1.361***
	(0.255)	(0.310)	(0.310)	(0.315)
romancatholic	0.921***	0.775**	0.776**	0.657*
	(0.328)	(0.371)	(0.377)	(0.385)
partyid	−0.171	−0.150	−0.151	−0.127
	(0.140)	(0.163)	(0.168)	(0.174)
secular.knowledge	−0.124	−0.250	−0.242	−0.252
	(0.314)	(0.366)	(0.473)	(0.477)
church		0.565**	0.569*	0.661*
		(0.271)	(0.326)	(0.349)
bible.collapsed		1.105**	1.104**	1.095**
		(0.511)	(0.512)	(0.517)
rps.ssm				0.162
				(0.732)
partyid:secular.knowledge	−0.059	−0.056	−0.055	−0.069
	(0.088)	(0.100)	(0.102)	(0.105)
church:bible.collapsed		−0.530***	−0.530***	−0.493***
		(0.179)	(0.179)	(0.179)
secular.knowledge:church			−0.004	0.020
			(0.142)	(0.146)
church:rps.ssm				−0.324
				(0.241)
Constant	1.898**	2.085*	2.074*	2.090*
	(0.839)	(1.095)	(1.174)	(1.202)
Observations	279	245	245	245
Log Likelihood	−138.275	−112.479	−112.478	−109.516
Akaike Inf. Crit.	298.551	252.957	254.957	253.033
Note:				*p<0.1; **p<0.05; ***p<0.01

Table 3: RPS does not promote cross-over issue constraint for Evangelical Protestants and Roman Catholics

	SSM Support	Abortion Support
	(1)	(2)
age	−0.023*	−0.019
	(0.013)	(0.013)
female	0.549	0.319
	(0.370)	(0.371)
black	−0.537	−0.150
	(0.605)	(0.629)
education.collapsed	0.125	−0.067
	(0.155)	(0.155)
income.collapsed	0.041	0.036
	(0.067)	(0.067)
ideology.collapsed	−0.868***	−1.361***
	(0.290)	(0.315)
romancatholic	1.128***	0.657*
	(0.390)	(0.385)
partyid	−0.179	−0.127
	(0.169)	(0.174)
secular.knowledge	−0.036	−0.252
	(0.493)	(0.477)
church	0.013	0.661*
	(0.336)	(0.349)
bible.collapsed	−0.411	1.095**
	(0.526)	(0.517)
rps.abortion	0.003	
	(0.758)	
rps.ssm		0.162
		(0.732)
partyid:secular.knowledge	−0.037	−0.069
	(0.102)	(0.105)
church:bible.collapsed	0.039	−0.493***
	(0.182)	(0.179)
secular.knowledge:church	0.014	0.020
	(0.145)	(0.146)
church:rps.abortion	−0.411	
	(0.251)	
church:rps.ssm		−0.324
		(0.241)
Constant	2.746**	2.090*
	(1.272)	(1.202)
Observations	246	245
Log Likelihood	−112.603	−109.516
Akaike Inf. Crit.	259.206	253.033
Note:		*p<0.1; **p<0.05; ***p<0.01

Table 4: Mainline Protestants' support of SSM Replication

	Dependent variable:			
	ssm			
	Secular	Naive	Culture Wars	RPS
	(1)	(2)	(3)	(4)
partyid	−0.182	−0.095	−0.080	−0.064
	(0.165)	(0.178)	(0.181)	(0.188)
secular.knowledge	0.911*	0.924*	0.646	0.579
	(0.499)	(0.508)	(0.696)	(0.718)
church		−0.365	−0.436	−0.455
		(0.439)	(0.459)	(0.482)
bible.collapsed		−0.130	0.010	−0.113
		(0.652)	(0.703)	(0.716)
rps.ssm				0.633
				(1.063)
partyid:secular.knowledge	−0.176	−0.229*	−0.236*	−0.241*
	(0.120)	(0.127)	(0.128)	(0.130)
church:bible.collapsed		0.140	0.083	0.108
		(0.275)	(0.294)	(0.301)
secular.knowledge:church			0.131	0.120
			(0.226)	(0.238)
church:rps.ssm				0.058
				(0.416)
Constant	0.509	0.905	1.000	0.715
	(0.624)	(1.103)	(1.125)	(1.153)
Observations	91	82	82	82
Log Likelihood	−52.919	−47.535	−47.366	−46.415
Akaike Inf. Crit.	113.837	109.070	110.731	112.829
Note:				*p<0.1; **p<0.05; ***p<0.01

Table 5: Mainline Protestants' support of Abortion Replication

	Dependent variable:			
	abortion			
	Secular	Naive	Culture Wars	RPS
	(1)	(2)	(3)	(4)
partyid	−0.309*	−0.290	−0.233	−0.239
	(0.174)	(0.191)	(0.200)	(0.203)
secular.knowledge	0.557	0.503	−0.754	−0.714
	(0.537)	(0.559)	(0.801)	(0.819)
church		−0.214	−0.607	−0.606
		(0.438)	(0.492)	(0.504)
bible.collapsed		0.025	0.814	0.850
		(0.679)	(0.842)	(0.856)
rps.ssm				−0.183
				(1.186)
partyid:secular.knowledge	−0.123	−0.153	−0.221	−0.226
	(0.125)	(0.134)	(0.146)	(0.147)
church:bible.collapsed		0.169	−0.106	−0.109
		(0.281)	(0.325)	(0.329)
secular.knowledge:church			0.632**	0.635**
			(0.262)	(0.266)
church:rps.ssm				−0.036
				(0.437)
Constant	1.338*	1.329	1.868	1.970
	(0.697)	(1.164)	(1.284)	(1.341)
Observations	93	84	84	84
Log Likelihood	−51.172	−44.908	−41.624	−41.520
Akaike Inf. Crit.	110.345	103.817	99.248	103.039
Note:		*p<0.1; **p<0.05; ***p<0.01		

Table 6: Beliefs about church teaching on same-sex marriage by personal opinion on same-sex marriage Replication

	Opposes	Supports
Church supports SSM	0.04	0.50
Church supports civil unions	0.25	0.15
Church opposes SSM	0.62	0.15
Doesn't know	0.10	0.20
Total	1.00	1.00
n	199	201

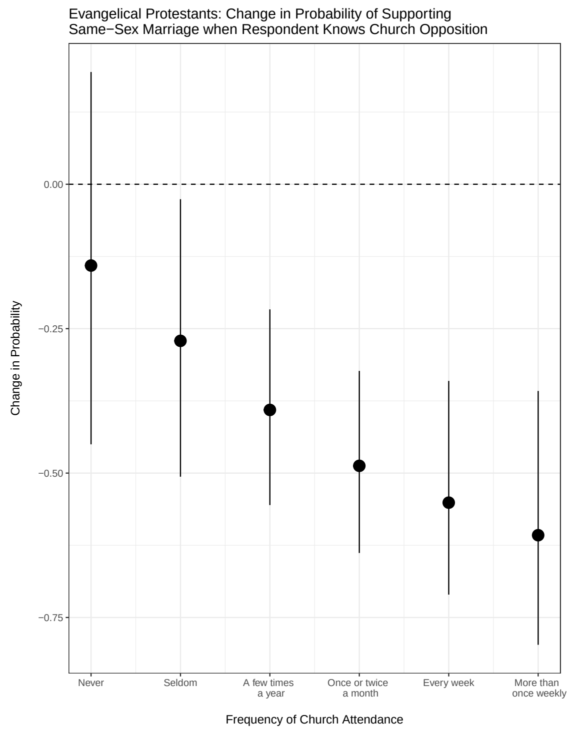
Table 7: Beliefs about church teaching on abortion by personal opinion on unrestricted abortion rights Replication

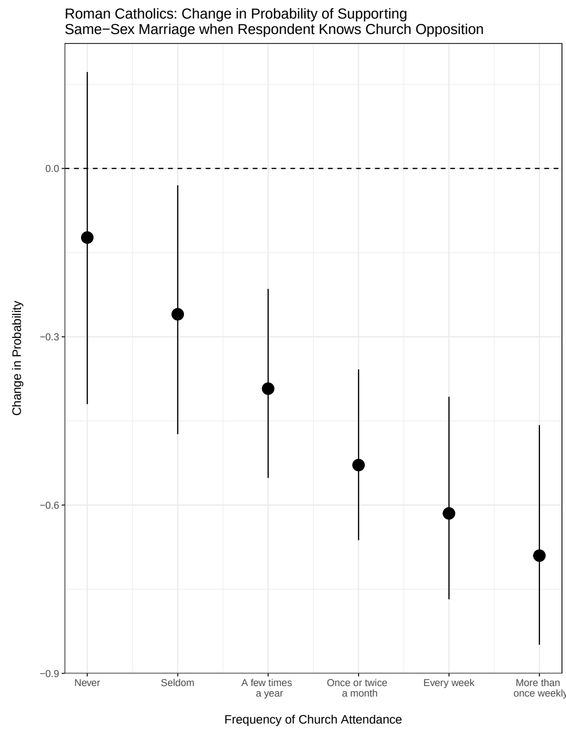
	0	1	NA
1	Always Permit	0	0.01
2	DK	0	0.06
3	Establish Need	0	0.08
4	Never Permit	0	0.39
5	Standard GOP Exceptions	0	0.45
6	Always Permit	1	0.34
7	DK	1	0.16
8	Establish Need	1	0.14
9	Never Permit	1	0.17
10	Standard GOP Exceptions	1	0.19

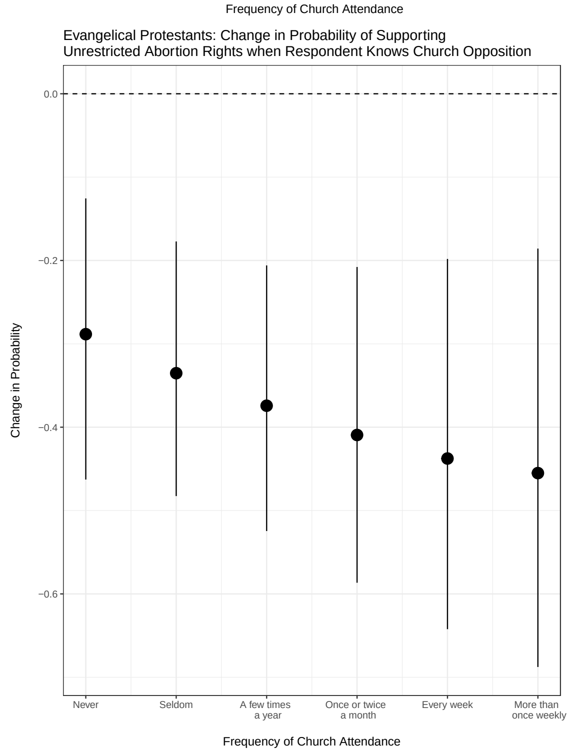
Table 8: Incorrect RPS reports only influence political attitudes through church attendance: support for same-sex marriage and unrestricted abortion rights, evangelical Protestants and Roman Catholics Replication

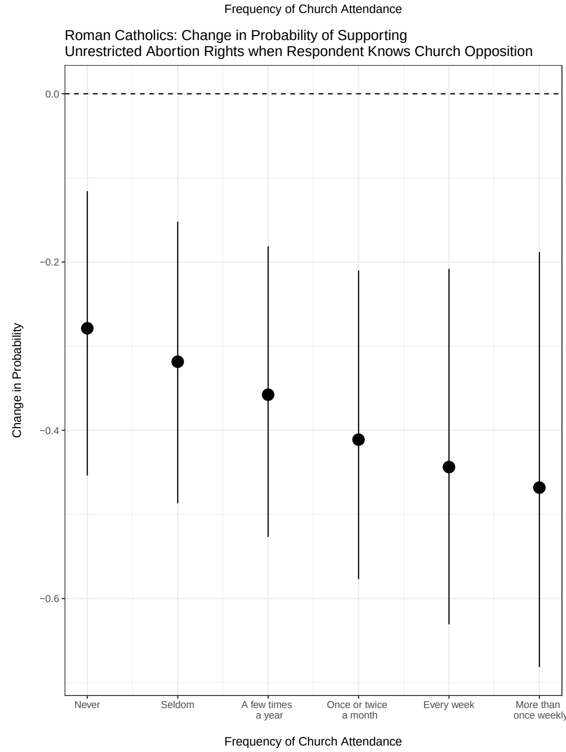
	Dependent variable:	
	ssm SSM Support	abortion Abortion Support
	(1)	(2)
age	−0.029**	−0.014
	(0.014)	(0.014)
female	0.686*	0.390
	(0.378)	(0.386)
black	−0.287	−0.705
	(0.614)	(0.678)
education.collapsed	0.048	0.008
	(0.155)	(0.168)
income.collapsed	0.049	0.050
	(0.068)	(0.069)
ideology.collapsed	−0.917***	−1.397***
	(0.304)	(0.324)
romancatholic	1.127***	0.493
	(0.393)	(0.423)
partyid	−0.236	−0.127
	(0.181)	(0.180)
secular.knowledge	−0.095	−0.202
	(0.500)	(0.489)
church	−0.096	0.309
	(0.338)	(0.361)
bible.collapsed	−0.123	1.028*
	(0.527)	(0.530)
incorrect.rps.ssm	0.275	
	(0.750)	
incorrect.rps.abortion		−0.734
		(0.828)
partyid:secular.knowledge	−0.012	−0.088
	(0.108)	(0.108)
church:bible.collapsed	−0.148	−0.472**
	(0.178)	(0.189)
secular.knowledge:church	−0.033	−0.039
	(0.148)	(0.156)
church:incorrect.rps.ssm	0.492*	
	(0.251)	
church:incorrect.rps.abortion		0.790***
		(0.287)
Constant	2.831**	2.192
	(1.343)	(1.342)
Observations	246	245
Log Likelihood	−106.152	−102.855
Akaike Inf. Crit.	246.303	239.709
Note:		*p<0.1; **p<0.05; ***p<0.01

4 Original Figures









5 My Extension

For the extension of this project, I decided to look at the "Culture Wars" model and remove the interaction term of secular knowledge and church attendance to test if this term contributes to the model's fit. I choose to do this because in almost every (except Mainline Protestant's support of abortion rights) this term is not statisically significant to the model. So, I looked at three out of the four "Culture Wars" logistic regression models that reported that the term was not statisically significant, removed the interaction term, then compared it to the original model with the term present using a partial F-test.

$H_0 : \beta_{\text{church}\times\text{secular.knowledge}} = 0$

$H_1 : \beta_{\text{church}\times\text{secular.knowledge}} \neq 0$

```
#Evangelical and Roman Catholics: SSM
cw1 <- glm(ssm ~ age + female + black + education.collapsed
           + income.collapsed + ideology.collapsed + romancatholic
           + partyid + secular.knowledge + partyid*secular.knowledge
           + church + bible.collapsed + church*bible.collapsed
           + church*secular.knowledge,
           data=evan.cath, family="binomial")
summary(cw1)

cw1_reduced <- glm(ssm ~ age + female + black + education.collapsed
           + income.collapsed + ideology.collapsed + romancatholic
           + partyid + secular.knowledge + partyid*secular.knowledge
           + church + bible.collapsed + church*bible.collapsed,
           data=evan.cath, family="binomial")
summary(cw1)

evan.cath_ssm <- anova(cw1_reduced, cw1, test = "Chisq")
```

Table 9: Partial F-test: Evangelical Protestants and Roman Catholics' support of SSM

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	232	237.110			
2	231	237.099	1	0.011	0.918

Table 10: Partial F-test: Evangelical Protestants and Roman Catholics' support of Abortion

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	231	224.957			
2	230	224.957	1	0.001	0.980

Table 11: Partial F-test: Mainline Protestants' support of Abortion

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	75	95.070			
2	74	94.731	1	0.339	0.561

Based on the p-values in each test (they are all greater than 0.05), we fail to reject the null hypothesis that the interaction term secular knowledge x church attendence has no effect on if a respondant supports an issue in the "Culture Wars" model. Therefore, for these particular models, it seems that this interaction is not needed.